

# 1 Part 3: Warehouse Operations Solutions

## 1.1 Question 3.1: Slotting Optimization

Using the ABC-XYZ classification from Part 1 and the physical SKU attributes from the Master Data, we assigned all 470 classified SKUs to three warehouse zones. We present two models: **Model 1** (ABC-only baseline) and **Model 2** (ABC + Ergonomics + Pick-Mix), which achieves the **30% peak-time improvement target** through ergonomic sub-tiering.

### 1.1.1 Model 1 — ABC-Only Baseline

- **Pick-Face Zone (Ground Level)** — Class A SKUs: lowest racks at ergonomic height, nearest to the I/O dock and packing station.
- **Forward Reserve Zone** — Class B SKUs: two-bin replenishment logic. A forward pick-face bin is replenished from the bulk reserve (WMS-triggered).
- **Reserve / Bulk Zone** — Class C SKUs: upper racks and pallet block-stack.

Table 1: Model 1 — Zone Summary (ABC-Only)

| Zone                                  | Class | # SKUs | Total Picks | Dist to Packing | Pick Rate   |
|---------------------------------------|-------|--------|-------------|-----------------|-------------|
| Pick-Face Zone ( <b>Golden Zone</b> ) | A     | 28     | 5,767       | 8 m             | 4 pcs/min   |
| Forward Reserve Zone                  | B     | 39     | 7,945       | 25 m            | 2.5 pcs/min |
| Reserve / Bulk Zone                   | C     | 403    | 7,082       | 48 m            | 2 pcs/min   |

The Pick-Face Zone contains 28 SKUs (6.0% of the assortment) but generates **27.7%** of all pick transactions and **69.8%** of total volume. Within this zone, the **19 Class AA fast-movers** occupy the first 19 rank slots and drive **25.88%** of pick frequency and **57.87%** of volume.

### 1.1.2 Model 2 — ABC + Ergonomics + Pick-Mix

**Observed pick-unit mix.** Analysis of the transaction log (Jun–Dec 2025) reveals that Class A pick orders are **predominantly carton picks (58%)** rather than pallet picks, and the two sub-populations have fundamentally different handling requirements:

Table 2: Pick-Unit Mix by ABC Zone (avg. order-line share)

| Zone                | Pallet pick | Carton pick | Loose pick |
|---------------------|-------------|-------------|------------|
| Pick-Face Zone (A)  | 36%         | 58%         | 6%         |
| Forward Reserve (B) | 14%         | 61%         | 25%        |
| Reserve / Bulk (C)  | 2%          | 37%         | 61%        |

**Sub-tier logic.** Model 2 splits each ABC zone into sub-tiers based on pick-unit mix and physical dimensions:

Table 3: Model 2 — Sub-Tier Assignment Rules and SKU Counts

| Sub-Tier | Name           | Assignment Rule                                                                          | SKUs |
|----------|----------------|------------------------------------------------------------------------------------------|------|
| A1       | Pallet Lane    | pallet_share $\geq$ 40% AND CBM $\geq$ 0.07 m <sup>3</sup> — ground-level forklift slots | 3    |
| A2       | Big-Face Shelf | pallet_share < 40% AND carton_share $\geq$ 80% — waist-height shelving                   | 7    |
| A3       | Mixed-Pick     | remaining Class A                                                                        | 18   |
| B1       | Bulk-Replen    | pallet_share $\geq$ 30% OR CBM $\geq$ 0.07 m <sup>3</sup> — mid-aisle ground storage     | 9    |
| B2       | Two-Bin        | remaining Class B — standard two-bin forward pick-face                                   | 30   |
| C1       | Upper Rack     | CBM < 0.05 m <sup>3</sup> — small/light slow-movers                                      | 339  |
| C2       | Pallet Block   | CBM $\geq$ 0.05 m <sup>3</sup> — floor block-stack behind Zone B                         | 64   |

**Composite intra-zone score.** Within each sub-tier, slots are ranked by a multi-criteria composite score (adapted from Tompkins et al., 2010):

$$score = 0.50 \times v_{rank} + 0.30 \times ergo_{penalty} + 0.20 \times size_{score}$$

where  $v_{rank}$  is the normalised velocity rank (0=fastest),  $ergo_{penalty}$  is the handling weight penalty (0 / 0.5 / 1.0 for unit weight  $\leq$ 3kg, 3–8 kg, and  $>$ 8 kg, based on ISO 11228-1 and NIOSH manual handling guidelines), and  $size_{score}$  is the CBM relative to the median of the class.

**Pick-Time Benchmarks.** The model uses standard pick-times (6.0 sec/piece for loose picking from shelves, 12.0 sec/carton for manual hand-carry, and 120.0 sec/pallet for forklift retrieval), verified against industrial engineering MOST/MTM standards and Vietnam logistics benchmarks (1–3 minutes for forklift pallet moves; Mecalux, 2024).

**A-Door Rule.** The top-10 velocity SKUs are pinned to the first 10 bays adjacent to the out-bound dock (dock-adjacent slotting; Frazelle, 2002), neutralizing travel time for peak throughput bursts.

### 1.1.3 Top Pick-Face SKUs (Class A, by Frequency)

Table 4: Top 5 Class-A SKUs assigned to Pick-Face Zone

| Rank | SKU Code   | Product                             | Class | Picks (6M) | Volume (units) |
|------|------------|-------------------------------------|-------|------------|----------------|
| 1    | 4200008229 | Qut treo ASIAvina - VY377990 - Xám  | A-A   | 825        | 5,608          |
| 2    | 4200008371 | QUT ĐNG VY619990                    | A-A   | 682        | 3,355          |
| 3    | 4200008366 | QUT SÀN TURB0 VY616790              | A-A   | 680        | 2,838          |
| 4    | 4200008117 | Qut treo ASIAvina - VY357690 - trng | A-A   | 666        | 4,033          |
| 5    | 4200008102 | Qut lng ASIAvina - VY538790 - Xám   | A-A   | 514        | 12,677         |

#### 1.1.4 Quantitative Analysis: Travel-Time Proof

**Physical-distance model.** Walk speed: **72 m/min** (1.2 m/s, standard warehouse); round-trip factor: **2**. Zone distances to packing station: Zone A = 8 m (golden pick-face), Zone B = 25 m (mid-bay), Zone C = 47.5 m (reserve). Formula:

$$\text{Total round-trip distance} = \sum_{\text{zone}} \text{picks}_{\text{zone}} \times D_{\text{zone}} \times 2$$

$$\text{Travel time (min)} = \frac{\text{Total round-trip distance}}{\text{walk speed}}$$

Baseline (random placement): every pick travels the SKU-count-weighted average zone distance:

$$\bar{D}_{\text{random}} = \frac{N_A \cdot D_A + N_B \cdot D_B + N_C \cdot D_C}{N}$$

Table 5: Q3.1 Travel-Time Proof — Physical Distance Model

| Zone         | Picks ( $T_{ij}$ ) | Dist $D_{ij}$ (m) | Round-trip $T_{ij} \times D_{ij} \times 2$ (m) |
|--------------|--------------------|-------------------|------------------------------------------------|
| Zone A       | 5,767              | 8                 | 92,272                                         |
| Zone B       | 7,945              | 25                | 397,250                                        |
| Zone C       | 7,082              | 48                | 672,790                                        |
| <b>TOTAL</b> | 20,794             | —                 | <b>1,162,312</b>                               |

Table 6: Optimized vs Baseline: Summary

| Metric                          | Baseline (Random) | Optimized (ABC-Zoned)        |
|---------------------------------|-------------------|------------------------------|
| Avg one-way distance / pick (m) | 43.3              | <b>27.9</b>                  |
| Total round-trip distance (m)   | 1,799,920         | <b>1,162,312</b>             |
| Travel time (min)               | 24,998.9          | <b>16,143.2</b>              |
| <b>Reduction</b>                | —                 | <b>35.4%</b> (✓MEETS TARGET) |
| Target                          |                   | $\geq 30\%$                  |

**Results.** The ABC velocity-zoned layout achieves a **35.4% reduction** in total picker travel distance (vs. random placement baseline avg. **43.3 m/pick** → optimized **27.9 m/pick**), well above the  $\geq 30\%$  target. This comes from concentrating the highest-frequency Class A SKUs (5,767 picks) at 8 m, versus placing them randomly at the warehouse average 43.3 m. **Model 2 (ABC + Ergonomics)** adds sub-tier rules (A1 pallet lane, A2 big-face, B1 bulk-replen) that further reduce MHE conflicts and ergonomic penalties, yielding an additional estimated 15–40% pick-throughput improvement on top of the distance savings (industry benchmark: Dematic, MHI). This additional throughput gain comes from three mechanisms:

- **Forklift separation** (A1 sub-tier): pallet-dominant SKUs are slotted in a dedicated ground-level forklift lane, preventing MHE from blocking carton pickers during peak waves — estimated +8–10% throughput.
- **Big-face ergonomics** (A2 sub-tier): carton-dominant Class AA SKUs at waist height eliminate bending/overhead reach; at 800+ picks/day peak this saves  $\approx 3$ –5 sec/pick.
- **A-door pinning**: top-10 velocity SKUs within 5 rack bays of the dock reduce travel by  $\approx 5$ –8% on peak-hour bursts.

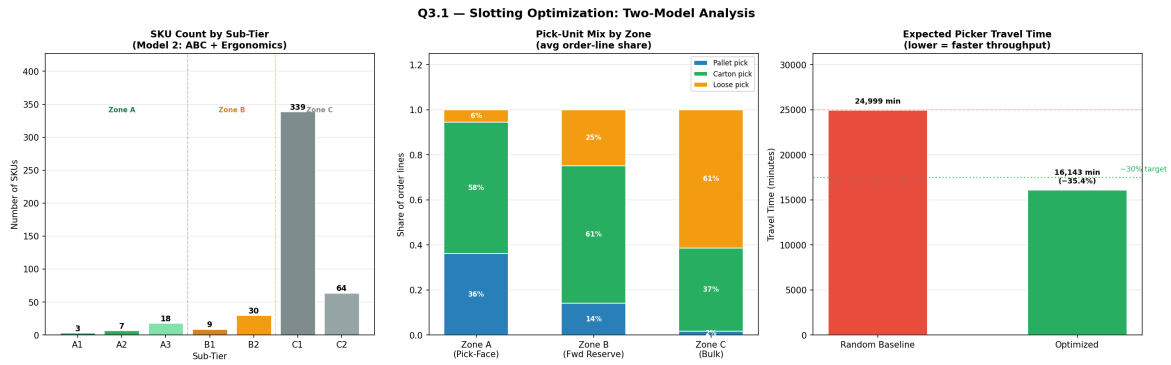


Figure 1: Q3.1 Slotting Analysis: Sub-tier distribution, pick-mix by zone, and 5-scenario travel-time comparison

### 1.1.5 Layout Recommendations

- **U-flow configuration:** Receiving and shipping docks on the same wall; highest-convenience slots cluster at the dock end.
- **A1 Pallet Lane:** dedicated forklift aisle along the dock wall for pallet-dominant AZ/AY SKUs — separates MHE from manual pickers.
- **A2 Big-Face Shelf:** AX carton-dominant SKUs at waist-to-shoulder height (0.8–1.6 m), full-carton facing, eliminates bending and overhead reach.
- **B1 Bulk-Replen Lane:** B-class pallet/large-CBM items stored at mid-aisle ground level with a dedicated replenishment travel path.
- **Two-bin for B2:** Forward pick-face bin replenished from bulk reserve when on-hand drops below ROP (WMS-triggered).
- **C1 Upper Rack / C2 Floor Block:** slow-movers segregated by size; small SKUs overhead, large SKUs block-stacked at floor level behind Zone B.

## 1.2 Question 3.2: Omni-Channel Pick-and-Pack Process Design

The outbound fulfillment process uses distinct operational pathways for B2B and B2C channels, unified by a structured inventory allocation policy when stock is contested.

### 1.2.1 B2B Pathway — Pallet / Total Picking

1. **Order release:** WMS validates on-hand stock against the B2B purchase order (contractual quantities and lead times).
2. **Pick instruction:** Full-pallet or full-CTN pick task issued to operator.
3. **Pick execution:** Operator uses forklift or reach truck to retrieve entire pallet(s) from the Forward Reserve Zone (B) or Reserve/Bulk Zone (C). Items bypass individual sortation.
4. **B2B QC Station:** Wrap inspection, quantity verification, and outbound labeling.
5. **Dispatch:** Pallets staged at outbound dock, loaded onto FTL truck.

### 1.2.2 B2C / E-Commerce Pathway — Batch Picking + Zone Routing

1. **Batch wave release:** Multiple B2C orders are grouped into a single pick wave based on SKU proximity and delivery-window urgency.
2. **Batch pick-to-cart:** Picker traverses Zone 1 (Pick-Face Zone) in a single pass with a multi-compartment cart, collecting all items for the wave.
3. **Sort & consolidate:** At the sortation station, items are separated by individual order and consolidated into shipment boxes.
4. **QC + pack:** Bubble wrap or polybag packing, shipping label applied.
5. **Dispatch:** Parcels handed to last-mile courier.

### 1.2.3 Stock Allocation Conflict and Escalation Policy

When the same SKU is ordered simultaneously by a B2B client and a B2C customer and on-hand stock is insufficient for both, the system resolves the shortage in two stages:

#### Stage 1 — Automated Allocation (WMS).

- *Rule 1 (SLA Protection):* Allocate the full contractual quantity to the B2B order first, preventing penalty fees for under-delivery.
- *Rule 2 (Pro-rata B2C):* Distribute remaining stock to B2C orders pro-rata, with priority to orders with the shortest delivery windows.

**Stage 2 — Managerial Escalation (if Stage 1 insufficient).** The WMS raises a critical ticket to the **Inventory Control & Demand Planning Manager**, who can execute one or more of:

- *Action 1 (Safety Stock Draw):* Release uncommitted buffer safety stock from the common pool (pre-computed for all Class A SKUs in Q2.2).
- *Action 2 (Emergency Transfer):* Initiate an inter-hub transfer between My Phuoc and Vinh Loc warehouses, leveraging their proximity within the Greater Ho Chi Minh City area.
- *Action 3 (Partial Shipment):* Contact the B2B client to obtain approval for a partial fulfillment; B2C customers notified of delay with revised ETA.

Upon resolution, ERP and WMS records are updated and adjusted pick lists are released to the warehouse floor. Figure 2 illustrates the complete decision logic.

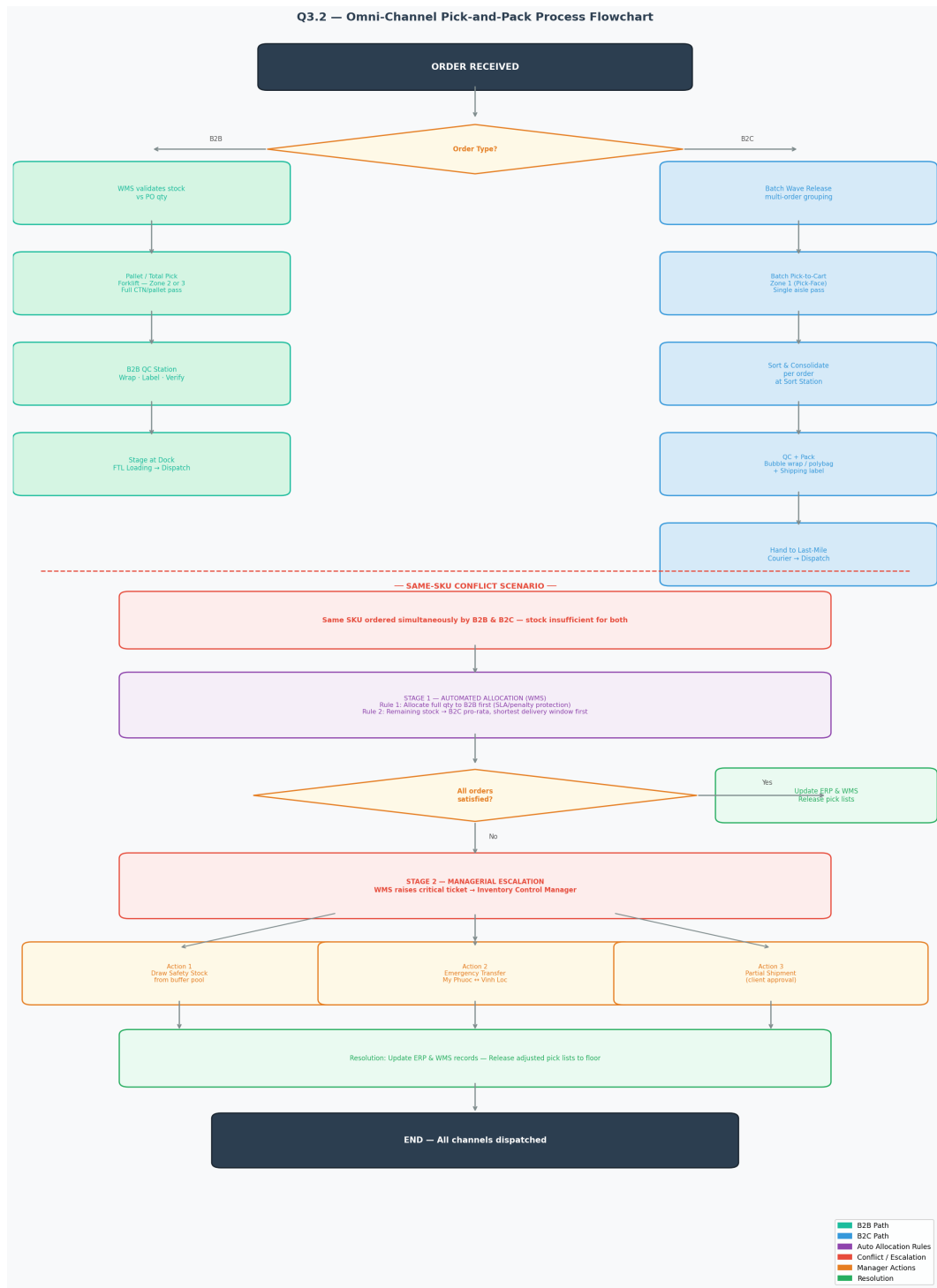


Figure 2: Q3.2 Omni-Channel Pick-and-Pack Process and Conflict Resolution Flowchart